Lieber Frank,

Dieses Dokument ist eine Zusammenfassung und Kollektion meines bisherigen Vorankommens. Ich beginne mit einer Einführung in die Diffenentialgeometrie, welche für dich nichts neues enthalten wird. Danach kommt eine kurze Einführung in die Morse Theorie. Hier ist denke ich folgendes für dich spannend:

1. Theorem 2.0.10. Hier führe ich den **refined Morse chart** ein, welcher neben der quadratischen Form von f auch die Metrik im Ursprung diagonalisiert.

2. Definition 2.0.16 definiert den Randoperator und danach definiere ich noch die Kohomologie.

Kapitel 4 ist denke ich im Gesamten spannend. Dort führe ich zunächst die verwendete Filtrierung ein und dann die Spektralsequenz. Der Beweis von Theorem 3.2.3 ist wohl der Höhepunkt. Ich habe auch zur Wiederholung in dem "Beweisidee" Container die Definition des verbindenden Homomorphismus der K-Theorie nochmal wiederholt. (Eine Einführung in die K-Theorie habe ich der Länge wegen weggelassen.) Der Beweis ist ziemlich technisch, hat zu diesem Zeitpunkt noch viele Lemma und ist auch noch nicht ganz fertig. Es fehlt noch die technisch saubere Ausarbeitung des letzten Arguments(das habe ich auch markiert). Hier wirst du denke ich viel wiederfinden von unseren Gesprächen.

Generell ist das hier mein *Arbeitsdokument* in etwas angepasster Form. Daher ist leider **noch** nicht alles einheitlich und etwas durcheinander und die hübschen Bilder fehlen noch. Auch möchte ich dich warnen, dass ich es noch nicht auf Rechtschreibung korrigiert habe und dank meiner Rechtschreibschwäche daher wohl noch viele Fehler zu finden sind.

Contents

1 Differentiable structures on topological manifolds

This section aims to set the playing field for the thesis. Due to the theory being of intrest for mathematicians and physisists alike there are many (equivalent) definitions used to introduce the objects. The main goal of this section is to avoid confusion, by giving the definitions instead relying on the reader to guess what a certain symbol stands for.

Definition 1.0.1 (Topological Manifold)**.** A second countable Hausdorffspace M is called **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, if for all $p \in M$ there exists an open neighborhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M and φ^{-1} a **local coordinate system around** p **on** M.

Definition 1.0.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension M.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$ the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$ we call the pair (M, c) a smooth manifold.

Corollary 1.0.3. Every transition functions φ_{ij} $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not just a homeomorphism but also a diffeomorphism due to $\varphi_{ji} = \varphi_{ij}^{-1}$

Definition 1.0.4. Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuos function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any one (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f \in U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 1.0.5. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be a open subset. With pointwise defined operations, the set $(\mathcal{E}(U),+,\cdot,\circ)$ becomes an $\mathbb{R}-$algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf as a reminder. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \mathrm{res}^U_V : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \,. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 1.0.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_w = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p \in U} \mathcal{E}(U) \right) \Big/ \sim_p \,.$$

For the set of germs and call it the **stalk in** p. Here $\sum$ denotes the co-product (also called sum) in **Top** and hence the disjoint union.

Corollary 1.0.7. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}-$algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \mathrm{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\mathrm{eval}_p)$. Hence, the pair $(M,\mathcal{E})$ gives us a locally ringed space.

Proof. Here, we only need to prove the statement about the locality of the stalks. This follows from $f_p \in \mathcal{E}_p(M)$ being invertible if and only if $f(p) \neq 0$ which is the same as $f_p \notin \ker(\mathrm{eval}_p)$. □

Definition 1.0.8. Let (M,c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibnitz-rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f,g \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$TM_p := \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})\,.$$

Corollary 1.0.9. Let (M,c) be a smooth manifold and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ $(\varphi \in \mathfrak{A} \in c)$. Then

$$\xi = \frac{\partial}{\partial x_j}\Big|_p : \mathcal{E}_p(M) \to \mathbb{R}$$
$$f_p \mapsto \xi(f_p) = \frac{\partial}{\partial x_j}\Big|_{x_0}(f \circ \varphi^{-1})$$

is well-defined and a tangent vector. In fact, the family

$$\left(\frac{\partial}{\partial x_1}\Big|_p, \dots, \frac{\partial}{\partial x_m}\Big|_p\right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

Definition 1.0.10. For a smooth manifold (M,c) the sum $\sum_p TM_p$ comes with a natural projection

$$\pi : TM \to M$$
$$\xi \mapsto p \text{ where } \xi \in TM_p$$

Furthermore, the **local vector fields** with respect to a chart $\varphi : U \to V$

$$\frac{\partial}{\partial x_j} : U \to \pi^{-1}(U)$$
$$p \mapsto \frac{\partial}{\partial x_j}\Big|_p$$

induce a local trivialization:

$$\pi^{-1}(U) \cong U \times \mathbb{R}^m .$$

We can induce a topology on TM such that all those trivializations are continuos. This then gives an atlas for TM such that we have a $2m$-dimensional manifold. To be precise, the atlas is given by the maps $\pi^{-1}(U_i) \to R^m \times R^m; x \mapsto (\pi(x), q_{\varphi\circ\pi(x)}(x))$ where q_p denotes the coordinate map corresponding to the basis $(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p)$ that depends on the chart φ_i. In fact, this yields a smooth manifold and a (smooth) vector bundle of dimension m. We call TM the **tangent bundle**.

Definition 1.0.11 (The Derivative)**.** Let (M,c) and (M',c') be smooth manifolds (from now on we suppress the differentiable structure in our notation). We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth function between the Tangent bundles as follows:

$$Df : TM \to TM' \ , Df_p(\xi)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi \in T_pM$, $g_p \in E_p(M')$.

Corollary 1.0.12. Let $f : M \to \mathbb{R}$ be smooth and $\varphi : U \to V$ be a chart. Then we can interprete $\mathrm{d}f$ as a one-form. To be prezice asume q to be the coordinate funktion $T\mathbb{R} \to \mathbb{R}$ from the basis induced by the identity as a chart. $v \in \Gamma TM$ we have

$$q \circ \mathrm{d}f(v) = v \cdot f$$

Here on the righd side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We keep this notations so vector fields can take funktions as an input.

Proof. We proof this by showing it for the section $\frac{\partial}{\partial x_i}$ and thereby for any, since those sections form a basis of the space of sections as a $C^\infty(M,\mathbb{R})$ vectorspace. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f(\frac{\partial}{\partial x_i}\Big|_p)(g_p) = \frac{\partial}{\partial x_i}\Big|_p(g \circ f)_p = \frac{\partial}{\partial x_i}\Big|_{x_0}(g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}\Big|_p(g_p) \cdot \frac{\partial}{\partial x_i}\Big|_p(f_p)$$

Hence, $\mathrm{d}f(v) = v(f)\frac{\partial}{\partial x}$ letting us conclude the statement. □

Definition 1.0.13 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(TM^* \otimes TM^*)$ into the tensor product of the dual Tangend bundle with itself is a **riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non degenerate**, meaning for a $v_p \in T_pM$ $g_p(v_p, w_p) = 0$ $\forall w_p \in T_pM$ then $v_p = 0$.
2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ $\forall v_p, w_p \in T_pM$.
3. g is **positiv definite**, meaning that $g_p(v_p, v_p) \geq 0 \forall v_p$ and vanishes only for $v_p = 0$.

We cal a touple (M, g) a **riemannian manifold**.

Definition 1.0.14. Let M, N, L be smooth manifolds and $f : M \times N \to M' \times N'$. Let $X \in \gamma(TM)$ and $Y \in \Gamma(TN)$. We define $\tilde{X} \in \Gamma(T(M \times N))$ as follows. Let $f_p \in \mathcal{E}_p(M, N)$

$$\tilde{X}(f_p) = X((f \circ i_M)_{p_M}) \quad \text{where } p_M = \pi_1(p)\,.$$

Definition 1.0.15 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vecor field V:

$$q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V)\,.$$

Here q denotes the canonical coordinate funtion $T\mathbb{R} \to \mathbb{R}$.

Definition 1.0.16. Let (M, g) be a riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be :

$$g_{ij} = g(\frac{\partial}{\partial x_i}, \frac{\partial}{\partial x_j})\,.$$

Then if two vectorfields v, w over U are given with $v = \sum v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$ we can calculate the metric locally:

$$g(v,w) = \sum_{ij} g_{ij} v^i w^j \; : U \to \mathbb{R}$$

Furthermore, we can invert the matricies $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure which can be seen by the construction via cramers rule) to get smooth functinos

$$\begin{aligned} g^{ij} &: U \to \mathbb{R} \\ &p \mapsto g^{ij}(p) = \left((g_{kl}(p))^{-1}_{kl}\right)_{ij} \end{aligned}$$

Lemma 1.0.17. *Let $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\operatorname{grad}(f) = \sum_{i,j} g^{ij} \left(\frac{\partial}{\partial x_i}(f)\right) \frac{\partial}{\partial x_j} =: \sum_{i,j} g^{ij} \frac{\partial f}{\partial x_i} \frac{\partial}{\partial x_j}\,.$$

Proof. Assume that $v, w \in \Gamma(TM)$ such that $v = \sum_i v^i \frac{\partial}{\partial x_i}$ and $w = \sum_i w^i \frac{\partial}{\partial x_i}$. Then $g(v,w) = \sum_{i,j} g_{ij} v^i w^i$. Suppose that $\operatorname{grad}(f) = \sum_j G^j \frac{\partial}{\partial x_j}$ and q is the coordinate map $T\mathbb{R} \to \mathbb{R}$ in each tangend space. Then by definition of the gradient we have:

$$\frac{\partial}{\partial x_j}(f) = q \circ \mathrm{d}f(\frac{\partial}{\partial x_j}) = q \circ \frac{\partial f}{\partial x_j} = g(\operatorname{grad}(f), \frac{\partial}{\partial x_j}) = \sum_{ij} g_{ij} G^i$$

Hence,

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right) \\ \Leftrightarrow (G^1, \cdots G^m) &= \left(\frac{\partial}{\partial x_1}(f), \cdots, \frac{\partial}{\partial x_m}(f)\right)(g^{ij})\,. \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \frac{\partial}{\partial x_i}(f)$.

□

Definition 1.0.18 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) := \pi_1(\Omega \cup \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t,p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t,p))) = \varphi(s+t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi : t(p) := \varphi(t, p)$ to keep the notation bearable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 1.0.19. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff},\ t \mapsto \varphi_t$$

is a homomorphism. To see the well definition notice how for all $t \in \mathbb{R}$ φ_{-t} is an inverse of φ_t.

Definition 1.0.20 (Vector Field of a Dynamical System)**.** Let φ be a dynamical system on a smooth manifold M. Then we call the mapping

$$p \mapsto X(p) := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \varphi_t(p) \in T_pM$$

the **associated vecor field**. This is in fact a smooth vector field.

Whilst differentiating leads to a smooth vector field, we can also go the other direktion by locally integrating a vector field to get a flow. To be precise: Those concepts are inverse to one another or in other words

hier genauer werden

Theorem 1.0.21. *Let M be a smooth manifold. Then there is a one to one correspondence between global maximal flows and vector fields.*

2 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [**banyaga2004lectures**]. In Morse Theory, we will always deal with asmooth funkition $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 2.0.1 (Kritical Points)**.** We call a point $p \in M$ **critical**, if $\mathrm{d}f|_p$ vanishes meaning that

$$\mathrm{d}f|_p : T_pM \to T_{f(p)}\mathbb{R}$$

is trivial. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 2.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth funktion. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear funktion

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangend vectors. Now choos two vectorfields Ξ and Z in a neighbourhoof of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. With this define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = (\Xi \cdot (Z \cdot f))\,(p)$$

Compare 1.0.12 for the notation.

Definition 2.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point.

The Hessian is a Symmetric bilinear and non-degnereate Map. For such a map there exists the following theorem:

Theorem 2.0.4 (Sylvesters Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that*

$$T^t \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^t \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^t \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^t \circ A \circ T') = k' + l' \,. \tag{1}$$

Hence it suffices to show that $k = k'$. Which we will do by proofing the claim:

$$k = \max\left\{\dim(U)|U \subseteq \mathbb{R}^n \text{subspace such that } x^t Ax > 0 \ \forall x \in U \setminus \{0\}\right\}$$

So we start by showing „$\leq$": Denote the first k colomus of T with $x_1, ..., x_k$. They form a basis of $\mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda i x_i$ we have that by bilinearity

$$x^t Ax = \sum_{i=1}^k \lambda_i x_i^t Ax = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^t Ax_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{2}$$

This conculdes the first inequality.

Now let U be any $k-$dimensional subspace such that for all non zero $x \in U$ $x^t Ax > 0$. By a calculation analog to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots x_n)$ the number $x^t Ax$ is less or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n-k)) - (n-k) + 0 = k\,. \tag{3}$$

This is the last inequality proving the statement. □

Corollary 2.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corrsponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)_{i,j}$$

Proof. This is just an easy calculation, since $\operatorname{Hess}_p(f)\left(\left.\frac{\partial}{\partial x_i}\right|_p, \left.\frac{\partial}{\partial x_j}\right|_p\right)$ is given by

$$\left(\left.\frac{\partial}{\partial x_i}\right|_p \left(\frac{\partial}{\partial x_j}(f)\right)_p\right) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\right)$$

□

The Hessian matrix $M_p(f)^i$ in two different charts $\varphi_i : U_i \to V_i$ for $i = 1, 2$ transforms via a congruence transformation by the Jacobian of the transition function If we have two different charts. Then the Matrix $M_p(f)$ of the Hessian transformes from one base to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = \mathrm{d}\varphi_{12}^t \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 2.0.6. By Sylvesters Law of interta, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is a invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each kritical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted with λ_p. If we cannot suppres the function without loss of clearity we will write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

Theorem 2.0.7 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse funktion $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ arround p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2$$

Proof. Compar for example Lemma3.11 in [**banyaga2004lectures**]. □

hier noch mehr Quellen!

Definition 2.0.8. Let $\varphi_t : M \to M$ be the local one-parameter group of diffeomorphisms generated by $-\mathrm{grad}(f)$, i.e. if $e : \mathbb{R} \to T\mathbb{R}$ denotes the canonical unit vector field that comes from the identity as the global chart on $\mathbb{R}$, we have that

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x) \circ e = -\mathrm{grad}(f)(\varphi_t(x)), \quad \text{and}$$
$$\varphi_0(x) = x .$$

We call that group the **gradient vevtor field** of f with respect to the gradient g. The integral curve $\gamma_x : (a, b) \to M$ given by

$$\gamma_x(t) = \varphi_t(x)$$

is called a gradient **flow line**.

Next we will deepen our **local** understanding of said vector fiel. Notice how the Morse Lemma is a statement about smooth manifolds-no metric needed nor mentioned. In fact there are many Morse charts and we can finde one, where the riemannian metric in p is diagonal with non-zero. To proof such a statement we first need a somewhat obvious lemma:

Lemma 2.0.9. *Assume that $\psi : U \to V$ is a chart and $L : R^m \to R^m$ a linear function. Denote the local coordinate maps of the Tangend bundle in a point p that comes from a basis in said point induced from a chart χ by q_χ. Then the diagramm*

$$\begin{tikzcd} T_pM \arrow[r, "q_\psi|_p"] \arrow[dr, "q_{L\circ\psi}|_p"'] & \mathbb{R}^m \arrow[d, "L"] \\ & \mathbb{R}^m \end{tikzcd}$$

commutes for all $p \in U$.

Proof. We show this by showing the inverse: For any $v \in \mathbb{R}^m$ we have the two $(q_\psi\big|_p)^{-1} \circ L^{-1}(v)$ and $(q_{\psi\circ L}|p)^{-1}$ and claim that they are the same. Assume that $L^{-1} = (\lambda_{ij})_{ij}$, $\psi(p) = x_0$, $L(x_0) = \tilde{x_0}$, $(v_1, \dots v_m) = v \in \mathbb{R}^m$ and denote the basis of T_pM coming from the chart ψ by $\left(\frac{\partial}{\partial x_i}\big|_p\right)_i$ and the one coming from $L^{-1} \circ \psi$ by $\left(\frac{\partial}{\partial \tilde{x}_i}\big|_p\right)_i$. We now calculate for $f_p \in \mathcal{E}_p M$:

$$\begin{aligned}
\left[(q_{\psi\circ L}\big|_p)^{-1}(v)\right](f_p) &= \sum_i v_i \frac{\partial}{\partial \tilde{x}_i}\Big|_p (f_p) \\
&= \sum_i v_i \frac{\partial}{\partial x_i}\Big|_{\tilde{x}_0} (f \circ \psi^{-1} \circ L^{-1}) \\
&= \sum_i v_i \sum_j \frac{\partial f \circ \psi^{-1}}{\partial x_j}\Big|_{x_0} \cdot \frac{\partial L_j^{-1}}{\partial x_i}\Big|_{\tilde{x_0}} \\
&= \sum_{i,j} v_i \frac{\partial}{\partial x_j}\Big|_p (f_p) \cdot \lambda_{ji} \\
&= \sum_{i,j} v_i q_\psi\big|_p^{-1}(e_j)(f_p) \cdot \lambda_{ji} \\
&= \left[q_\psi\big|_p^{-1}\big(\sum_{i,j} v_i \lambda_{ji} e^j\big)\right](f_p) \\
&= \left[q_\psi\big|_p^{-1}(L^{-1}(v))\right](f_p)
\end{aligned}$$

This concludes the proof. $\square$

Now we can proof that a Morse chart exists where the gradient is of particular easy form:

Theorem 2.0.10 (Simultaneous Diagonalisation of Quadratic Forms)**.** *Let p be a critical point of a Morse function f on a manifold M and let $g(p)$ be the Riemannian metric on T_pM. Then there exists a Morse chart ψ arround p such that the representation of $g(p)$ in the basis induced by the Morse chart is given by a a diagonal matrix $diag(\mu_1, \dots \mu_m)$ with $\mu_i > 0$ for all i .*

Proof. The quadratic form of $f \circ \psi^{-1} - f(p)$ in the coordinates of any Morse chart ψ is $q_H(v) = v^T H v$. The quadratic form induced by the Riemannian metric $g(p)$ is $q_G(v) = v^T G v$, where $v \in \mathbb{R}^m$ are the coordinate vectors with respect to the Morse chart. To be precice this means for $v, w \in \mathbb{R}^m$:

$$g \circ (q_\psi\big|_p \oplus q_\psi\big|_p)(v, w) = v^T G w \text{ and } f \circ \psi^{-1}(v) = f(p) + v^T H v\,.$$

We now aim to manipulate ψ such that G becomes diagonal: Since $g(p)$ is positive definite, G is also positive definite, and H is symmetric.

Consider the generalized eigenvalue problem $Hv = \lambda G v$. Since H and G are real symmetric matrices and G is positive definite, this problem has m real eigenvalues $\lambda_1, \dots, \lambda_m$ and corresponding eigenvectors $v_1, \dots, v_m$, which can be chosen to be orthogonal with respect to the bilinear form defined by G, such that $v_i^T G v_j = \delta_{ij}$, since if v_i and v_j are different eigenvectors with respect to different eigenvalues, we can calculate:

$$v_j^T H v_i = (Hv_j)^T v_i = \lambda_j (Gv_j)^T v_i = \lambda_j v_j^T G(v_i) \quad \text{and} \quad v_j^T H v_i = v_j^T \lambda_i G(v_i) = \lambda_i v_j^T G(v_i) .$$

Hence we have the equality

$$\lambda_j (Gv_j)^T v_i = \lambda_i v_j^T G(v_i) \Leftrightarrow \underbrace{(\lambda_j - \lambda_i)}_{\neq 0} v_j^T G(v_i) = 0$$

Hence all eigenspaces are orthogonal and we can use Gram Schmitt to make the bases of the eigenspaces orthogonal and by rescaling orthonormal.

Let $L = [v_1 | \dots | v_m]$ be the matrix whose columns are these G-orthonormal eigenvectors. The linear change of coordinates $y = Lz$ leads to new coordinates z. In these new coordinates, the quadratic forms transform as follows:

$$\begin{aligned} q_G(y) &= y^T G y = (Lz)^T G(Lz) = z^T L^T G L z = z^T I z = \sum_{i=1}^m z_i^2 \\ q_H(y) &= y^T H y = (Lz)^T H(Lz) = z^T L^T H L z \end{aligned}$$

Since $Hv_i = \lambda_i G v_i$, we have $L^T H L = \operatorname{diag}(\lambda_1, \dots, \lambda_m)$. The eigenvalues λ_i are real and have the same signature as the eigenvalues of H (l negative, $m - l$ positive).This is true to sysvesters law of inertia. By a further scaling of the v_i and a reordering the matrix $L^T H L$ can be brought into the form $\operatorname{diag}(-1, \dots, -1, 1, \dots, 1)$. (by doing this however, the matrix $L^T G L$ becomes $L^T G L = \operatorname{diag}(\mu_1, \dots \mu_m)$ with $\mu_j > 0 \ \forall j$). If ψ is the Morse chart we started with, then $L^{-1} \circ \psi$ is the new Morse chart:

$$f \circ (\psi^{-1} \circ L)(y) = f \circ \psi^{-1}(L(y)) = q_H(L(y) = \sum_{i=1}^l -y_i^2 + \sum_{i=l+1}^m y_i^1$$

Furthermore, we want to inspect the metric $g\big|_p : T_p M^2 \to \mathbb{R}$ with respect to the chart $L^{-1} \circ \psi$-induced basis $\left(\frac{\partial}{\partial x_1}\big|_p, \dots, \frac{\partial}{\partial x_m}\big|_p \right)$. By the lemma 2.0.9

$$g(q_{L^{-1} \circ \psi}\big|_p^{-1}, q_{L^{-1} \circ \psi}\big|_p^{-1})(v, w) = g(q_\psi\big|_p^{-1}, q_\psi\big|_p^{-1})(Lv, Lw) = (Lv)^T G L W = v^T L^T G L w$$

□

Definition 2.0.11. We call a Morse chart where the Matrix corresponding to the gradient in the critical point is of diagonal form a **refined Morse chart**.

Using such a chart, we can start looking at the flow itself and try to find easy local formulars around critical points. For this we start by looking at the Jacobian of the gradient in a critical point:

Definition 2.0.12 (Jacobian of the Gradient)**.** The gradient is a section into the tangent bundle, $\mathrm{grad}(f) : M \to TM$. Let $\psi : M \supseteq U \to V \subseteq \mathbb{R}^m$ be a chart. The coordinate map q_ψ on TU assigns to a vector $v \in T_pM$ (where $p \in U$ with $\psi(p) = x$) its components with respect to the basis $\left(\frac{\partial}{\partial x_i}\big|_p\right)$, i.e., $q_\psi(v) = (v_1, \dots, v_m)$ if $v = \sum_{i=1}^m v_i \frac{\partial}{\partial x_i}\big|_p$. We define the Jacobian of the gradient with respect to the chart ψ as the Jacobian matrix of the coordinate representation of the gradient in this chart:

$$J(\mathrm{grad}(f))_\psi(x) := J\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)(x) = \left(\frac{\partial(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1})_i}{\partial x_j}(x)\right)_{ij}$$

Lemma 2.0.13. *Let ψ be a coordinate system arround a critical point p. I.e. $\psi : p \in U \to V$ with $\psi(p) = 0$ and let $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$ be the induced basis of T_xM Then the Jacobian of the gradient reads*

$$J(\mathrm{grad}(f))_\psi(0) = \left(\sum_k g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij}. \tag{4}$$

Proof. Let q_ψ denote the coordinate funktion $TU \to R^m$ induced from the basis $\left(\frac{\partial}{\partial x_1}\big|_x, \dots, \frac{\partial}{\partial x_m}\big|_x\right)$. then the gradient in ψ reads:

$$q_\psi \circ \mathrm{grad}(f) = \left(\sum_k g^{k1} \frac{\partial f}{\partial x_k}, \dots, \sum_k g^{km} \frac{\partial f}{\partial x_k}\right)$$

Hence, we can differentiate:

$$\begin{aligned}
\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1} &= \left(\frac{\partial}{\partial x_j} \sum_k g^{ki} \frac{\partial f \circ \psi^{-1}}{\partial x_k}\right)_{ij} \\
&= \left(\sum_k \left[\frac{\partial g^{ki}}{\partial x_j} \frac{\partial f \circ \psi^{-1}}{\partial x_k} + g^{ki} \frac{\partial^2 f \circ \psi}{\partial x_j\, \partial x_k}\right]\right)_{ij}.
\end{aligned}$$

and now in $p = \psi^{-1}(0)$ we have

$$\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\Big|_0 = \left(\sum_k \left[g^{ki}(0) \frac{\partial^2 f \circ \psi^{-1}}{\partial x_j\, \partial x_k}\right]\right)_{ij}.$$

□

Now that we now how the gradient locally looks like ist reasonable to assume, that the flow locally looks like its exponential:

Lemma 2.0.14. *Let $t \in \mathbb{R}$ be small enough and $\varphi_t : M \to M$ be the flow map corresponding to the negative gradient. Assume that U is the domain of a chart ψ and $p \in U$ such that $\varphi_t(p) \in U$. Then the linear map $\mathrm{d}\varphi_t\big|_p : T_pM \to T_{\varphi_t(p)}M$ has a local representation in the coordinates induced by ψ given by:*

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\left(\sum_k g^{ki}(0)\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k}\right)_{ij} \cdot t\right).$$

Proof. For this we consider the function $(t, x) \mapsto \psi \circ \varphi_t(\psi^{-1}(x)) : \mathbb{R} \times U \to U$ For fixed x we can calculate

$$q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) = -q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x)))$$

and by changing the order of integration we have:

$$\begin{aligned}
\frac{\partial}{\partial t}\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) &= \frac{\partial}{\partial x}\frac{\partial}{\partial t}\psi \circ \varphi_t(\psi^{-1}(x)) \\
&= \frac{\partial}{\partial x} q_\psi \frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\psi^{-1}(x)) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)(\varphi_t(\psi^{-1}(x))) \\
&= -\frac{\partial}{\partial x} q_\psi \circ \mathrm{grad}(f)\left(\psi^{-1} \circ \psi \circ \varphi_t(\psi^{-1}(x))\right) \\
&= -\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right)\frac{\partial}{\partial x}\left(\psi \circ \varphi_t(\psi^{-1}(x))\right)
\end{aligned}$$

Hence by the theory of linear differential equation with $\frac{\partial}{\partial x}\psi \circ \varphi_0(\psi^{-1}(x)) = \mathbb{1}_m$ we have:

$$\frac{\partial}{\partial x}\psi \circ \varphi_t(\psi^{-1}(x)) = \exp\left(-\frac{\partial}{\partial x}\left(q_\psi \circ \mathrm{grad}(f) \circ \psi^{-1}\right) \cdot t\right).$$

The preceeding lemma now finishes the proof.

□

Hence in a refind Morse chart we have:

Remark 2.0.15. Let ψ be a refined Morse chart, and hence:

(a) $g^{ik}(0) = \delta_{ik}\mu_k$ with $\mu_k > 0$ for any k.

(b) $\frac{\partial^2 f \circ \psi^{-1}}{\partial x_j \partial x_k} = s_k 2\delta_{jk}$ where $s_k = -1$ for $k \leq \lambda_p$ and $s_k = 1$ else.

Then the Jacobian of the flow in the critical point p reads:

$$\begin{aligned}\frac{\partial}{\partial x}\psi\circ\varphi_t(\psi^{-1}(x)) &= \exp(-J(\mathrm{grad}(f))(p)\cdot t)\\ &= \exp\left(-\sum_k \underbrace{g^{ki}}_{=\delta_{ik}\mu_k}(0)\underbrace{\frac{\partial^2 f\circ\psi^{-1}}{\partial x_j\partial x_k}}_{=s_k 2\delta_{jk}}\cdot t\right)\\ &= \mathrm{diag}\left(e^{2\mu_1 t},\dots,e^{2\mu_{\lambda_p}t},-e^{2\mu_{\lambda_p+1}t},\dots,-e^{2\mu_m t}\right)\end{aligned}$$

we now skip a lot of theory that essentally uses the (un-)stable manifold theorem and the λ-lemma, to give the definition of the Morse chain complex for a Morse datum.

Definition 2.0.16 (Morse Homology)**.** Let $f : M \to \mathbb{R}$ be a Morse-Smale function on a Riemannian manifold (M, g). Furthermore let $\mathfrak{o}$ be an choice of orientations for all unstable Tangendspaces ($T_q^u M$ for all $q \in \mathrm{Crit}(f)$). The tuple $\mathfrak{A} := (g, f, \mathfrak{o})$ is a **Morse datum**. With this we define now the **Morse chain groups** for a given commutative ring with one (Λ):

$$C_k(M,\mathfrak{A},\Lambda) := \bigoplus_{\mathrm{Crit}(f)}\Lambda$$

To define a boundary operator between the chain groups on the generators we count flow lines between them. Lets say that p is a critical point of index k and q one of index $k+1$. Then we have the set of flow lines from q to p denoted by $\mathcal{M}(q \to p)$. For each $\gamma \in \mathcal{M}(q \to p)$ we assign a sign $n_\gamma \in \{\pm 1\}$. Let $x \in \gamma$. Then the tangent space in x splits $T_x W(q \to) = N_x(\gamma) \oplus \langle -\mathrm{grad}(f)(x)\rangle$. Furthermore, we can parallel transport $N_x(\gamma)$ to $T_p^u M$. Hence, the orientation in $T_q^u M$ can be compared to the one in $T_p^u M$ and the sign n_γ can be chosen accordingly.The **boundary operator** on a generator is defined as:

$$\partial_{k+1}(q) := \sum_{p\in\mathrm{Crit}_k(f)}\ \sum_{\gamma\in\mathcal{M}(q\to p)} n_\gamma$$

This gives rise to a chain complex and hence gives rise to a homology theory, and a cohomology theory by applying the hom functor to the chain complex.

Here i write a little bit about the Morse cohomology in german

Theorem 2.0.17 (Morse Cohomology)**.** *Um die Morse Kohomologie zu erhalten, dualizieren wir den Kettenkomplex für einen Ring G mit 1. Sei* $\mathrm{Crit}_k = p_1, ...p_l$ *eine Basis der $k-$ten Kettengruppe. Für*

$$\begin{aligned}\eta_i : C_k &\to G\\ p_j &\mapsto \begin{cases}0 & \textit{if } j = i\\ 1 & \textit{else}\end{cases}\end{aligned}$$

bekommen wir eine Basis $(\eta_1, ..., \eta_l)$ *der Cocetten Gruppe. Die Randabbilsung ist nun der spannende Teil. Seien* $(q_1, ..., q_r)$ *kritische Punkte von Index* $k+1$ *und* $(\xi_j)_{j=1,...,r}$ *die resultierende Basis von*C^{k+1}. *Dann*

$$\begin{aligned} \mathrm{d}_k : C^k &\to C^{k+1} \\ \eta_i &\mapsto \mathrm{d}\eta_i = \eta_i \circ \partial_{k+1} = \sum_j n_f(q_j, p_i)\xi_j. \end{aligned}$$

Hier beschreibt $n_f(q,p)$ *das zählen von Flusslionien von* q *nach* p *mittels* f.*Indem wir kritische Punkte und die formen identifizieren erhalten wird das folgende:*

$$\begin{aligned} C^k &= \bigoplus_{q \in \mathrm{Crit}_k} \mathbb{Z} = C_k. \\ \mathrm{d} : \ & C^k \to C^{k+1} \\ & q \mapsto \sum_{p \in \mathrm{Crit}_{k+1}} n_f(q,p) = \sum_{p \in \mathrm{Crit}_{k+1}} n_{-f}(p,q) = \partial_{-f}. \end{aligned}$$

Hier bezeichnet ∂_{-f} *den Randoperator* $C(M, -f, g)$.

Theorem 2.0.18 (Poincaré Dualität)**.** *Schnell sehen wir, dass*

$$C^k(M,f) = C_{m-k}(M,-f)$$

damit erhalten wir:

$$H^k(M) \cong H^k(M,f) = \ker(\partial_{-f}) / \mathrm{im}\partial_{-f} = H_{m-k}(M,-f) \cong H_k(M)$$

3 Morse-Theoretic Atiyah-Hirzebruch Spectralsequence

3.1 Conley index

This is just a short summary. Prooves are omitted and the goal is to remaind ourself of the definitinos.

Definition 3.1.1 (Compact invariant isolated subset)**.** For a flow $\varphi_t : M \to M$ on a locally compact metric space we call a subset $S \subseteq M$ **invariant subspace** if and only if $\varphi_t(S) = S$ for all $t \in \mathbb{R}$. For any subset $N \subseteq M$ we define the **maximal invariant subset**

$$\begin{aligned} I(N) &= \{x \in N \mid \varphi_t(x) \in N \forall t \in \mathbb{R}\} \\ &= \bigcap_{t\in\mathbb{R}} \varphi_t(N). \end{aligned}$$

A **compact invariant subset** S is called **isolated**, if a compact neighborhood N exists, such that $I(N) = S$.

Definition 3.1.2 (Index pairs)**.** Let S be an isolated compact invariant subset. A topological pair (N, L) of compact subsets of M where $L \subseteq N$ is called an **index pair of** S, if it satisfies the following:

1. $S = I(\overline{N \setminus L}) \subseteq (\overset{\circ}{N \setminus L})$.
2. $x \in L$ and $\varphi_{[0,t]}(x) \subseteq N$ implies that $\varphi_{[0,t]}(x) \subseteq L$. We call L **positively invariant in** N. Here $\varphi_{[0,t]}(x) := \{\varphi_{\tilde{t}}(x) | \tilde{t} \in [0,t]\}$.
3. For all $x \in N$ such that a t exists, with $\varphi_t(x) \notin N$, there exists a t' with $\varphi_{[0,t']}(x) \subseteq N$ and $\varphi_{t'}(x) \in L$.

This definition captures how the flow lines leave the invariant space S. Due to the third property we call L the **exit set**.

Definition 3.1.3 (Regular index pairs)**.** We call an index pair (N, L) **regular** if and only if the inclusion $I : L \hookrightarrow N$ is a cofibration.

Theorem 3.1.4 (Homotopy equivalence of index pairs)**.** *If S is an isolated compact invariant set and (N, L) and $(\tilde{N}, \tilde{L})$ are two index pairs of S. Then N/L and $\tilde{N}/\tilde{L}$ are homotopy equivalent as pointed spaces.*

Remark 3.1.5 (Critical points as isolated compact invariant subsets)**.** Let $f : M \to \mathbb{R}$ be a Morse function on a smooth Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$. Around q we have Morse charts $\varphi : U \to T_qM$ (after identifying q_i and ∂q_i) where $\varphi(W(q \to) \cap U) \subseteq T_q^u M$ and $\varphi(W(\to q) \cap U) \subseteq T_q^s M$. With this we can define the balls:

$$\begin{aligned} D_\varepsilon^s &= \{v \in T_q^s M | \, \|v\| \le \varepsilon\}, \\ D_\varepsilon^u &= \{v \in T_q^u M | \, \|v\| \le \varepsilon\}. \end{aligned}$$

They give rise to the index pair $N_q := \varphi^{-1}(D^s \times D^u)$ and $L_q = \varphi^{-1}(D^s \times \partial D^u)$. This index pair is in fact regular, since $(N_q, L_q) \cong (D^s \times D^u, D^s \times \partial D^u) \cong (D^u_\varepsilon, \partial D^u_\varepsilon)$. Now we know the K-group of such a pair:

$$K^i(N_q, L_q) = K^i(D^k, \partial D^k) = \begin{cases} \mathbb{Z} & \text{if } i - k \in 2\mathbb{N}, \\ 0 & \text{else.} \end{cases} \tag{5}$$

Notice that for $k = 0$ we need to consider $\partial D^k = \varnothing$, to get an index pair. That's why we let ∂ denote the manifold boundary insted of the topological boundary. However, now we can identify for all k:

$$C^k(M, \mathfrak{A}, \tilde{K}^l(pt) = \bigoplus_{\mathrm{Crit}_k(f)} \mathbb{Z} \cong \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q; \mathbb{Z}). \tag{6}$$

Furthermore, this isomorphism can be made canonically, by using orientations: because we get canonically induced generators by orientations of the unstable manifolds.

Example 3.1.6 (A map of homology groups)**.** Let $f : M \to \mathbb{R}$ be a Morse Smale function on a smooth compact Riemannian manifold (M, g). Let $q \in \mathrm{Crit}_k(f)$ and $p \in \mathrm{Crit}_{k-1}(f)$. Assume for a moment we already have an isolated regular compact index pair (N_2, N_0) of $S = W(p \to q) \cup \{p, q\}$. For a $c \in (f(p), f(q))$ we set $N_1 = N_0 \cup (N_2 \cap M^c)$, where M^c is the sublevel set. Clearly now (N_2, N_1) is an index pair for q and (N_1, N_0) is a pair for p. Assuming we have regular index pairs $(N_q.L_q)$ and (N_p, L_p) for p, q we can now define a map:

$$\Delta^k(q \to p) : K^{k-1}(N_p, L_p) \to K^k(N_q, L_q)$$

as the composition:

$$K^{k-1}(N_p, L_p; \mathbb{Z}) \xrightarrow{\cong} K^{k-1}(N_1, N_0; \mathbb{Z}) \xrightarrow{\delta_{triple}} K^k(N_2, N_1; \mathbb{Z}) \xrightarrow{\cong} K^k(N_q, L_q; \mathbb{Z})$$

where δ_{triple} is the connecting homomorphism, and the other two maps are induced by the homotopic equivalence of two index pairs corresponding to the same isolated compact invariant subset. Putting all together we can define a morphism:

$$\Delta^k : \bigoplus_{p \in \mathrm{Crit}_{k-1}(f)} K^{k-1}(N_p, L_p) \to \bigoplus_{q \in \mathrm{Crit}_k(f)} K^k(N_q, L_q) \tag{7}$$

Definition 3.1.7 (Regular index pairs and a filtration on M)**.** As always let $f : M \to \mathbb{R}$ be a Morse-Smale function on a compact smooth Riemannian manifold (M, g) of dimension m. Let $\varphi_t : M \to M$ be the flow given by $-\mathrm{grad} f$. For $0 \leq j \leq k \leq m$ define:

$$W(k, j) = \bigcup_{j \leq \lambda_p \leq \lambda_q \leq k} W(q \to p).$$

We know that this space is compact. Assume that N is a compact neighbourhood of reference

$W(k,j)$, such that $\mathrm{Crit}(f) \cap N = \mathrm{Crit}(f) \cap W(k,j)$, meaning that N does not contain any new critical points. Then the biggest invariant subspace from definition 3.1.2 is

$$I(N) := \{x \in N | \varphi_t(x) \in N \forall t \in \mathbb{R}\} = W(j,k).$$

This follows from every gradient flow line starting and ending in a critical point. Therefore, $W(k,j)$ is an isolated compact invariant set. By a corollary of the lambda lemma we know that

reference

$$\begin{aligned} W_j^s &:= \bigcup_{j \leq \lambda_p} W(\to p) \\ W_j^u &:= \bigcup_{\lambda_p \leq j} W(p \to) \end{aligned}$$

for all $j = 0, \ldots, m$ are compact as they are finite unions of compact sets. Now we define $N_m := M$ and choose a cofibered compact neighbourhood N_{m-1} of $W_{m-1}^{p\to}$ that is positively invariant and satisfies $N_{m-1} \cap W_m^{\to p} = \emptyset$ One could for example define:

$$N_{m-1} := N_m \setminus \Big(\bigcup_{q \in \mathrm{Crit}_m(f)} \mathring{N}_q \Big),$$

where N_q is taken from example 3.1.5. Once we check that N_{m-1} is an exit set with respect to N_m we can conclude that (N_m, N_{m-1}) is a regular index pair for $\mathrm{Crit}_m(f)$. The first statement is clear, since every gradient line starting in $N_m \setminus N_{m-1}$ has to pass through N_{m-1}, as they go towards other critical points. This tells us that they have to leave $N_m \setminus N_{m-1}$, which is enclosed by N_{m-1}. The second thing to check can be done by showing that there is a neighbourhood $U \subset N_M$ of N_{m-1} that deforms to N_{m-1}. For this notice that each N_q with $\lambda_q = m$ is homeomorphic to D^m. Now choose a open neighbourhood U_q in the disc of its boundary. By definition this retracts to the boundary with a retraction induced by the flow. And finally define $U := N_{m-1} \cap_{q \in \mathrm{Crit}_m(f)} U_q$. This retracts to N_{m-1} telling us that our index pair is indeed regular.

Now we want to inductively define a filtration: For this we choose a compact cofibered neighborhood N_{m-2} of W_{m-2}^u that is positifely invariant in N_{m-1} and has an empty intersection with $W_{m-1}^{\to p}$. This can be done similar to the construction of N_{m-1} but instead of cutting out neighbourhoods if $q \in \mathrm{Crit}_m(f)$, we can cut out tubular neighborhoods of W_{m-1}^s. By iterating this process we get a filtration:

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$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M, \tag{8}$$

such that (N_k, N_{j-1}) is a regular index pair for $W(k,j)$ for all $0 \leq j \leq k \leq m$.

3.2 The Spectral sequence

Definition 3.2.1. Let M be a smooth manifold with Morse data and

$$\emptyset =: N_{-1} \subset N_0 \subset N_1 \subset \cdots \subset N_{m-1} \subset N_m = M,$$

be a filtrations constructed in 3.1.7. Then we define the Groups

$$E_r^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]}{\ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_p, N_{p-r})\right]}.$$

To define a boundary operator, we proceed as follows: The following diagram commutes (by naturality of the boundary in the triple sequence).

$$\begin{tikzcd}
K^{p+q}(N_{p+r-1}, N_{p-r}) \arrow[r, "\alpha"] \arrow[d, "\delta^*_{triple}"] & K^{p+q}(N_p, N_{p-r}) \arrow[d, "\delta^*_{triple}"] & \\
K^{p+q+1}(N_{p+2r-1}, N_{p+r-1}) \arrow[r, "\beta"] & K^{p+q+1}(N_{p+2r-1}, N_p) \arrow[r] & K^{p+q+1}(N_{p+r-1}, N_p)
\end{tikzcd}$$

Hence, $\beta \circ \delta^*_{triple}$ factors over $K^{p+q}(N_{p+r-1}, N_{p-r})/\ker\alpha$ and because the bottom row is exact (by the triple sequence) we have get a map

$$K^{p+q}(N_{p+r-1}, N_{p-r})/\ker\alpha \to \ker\left[K_{p+q+1}(N_{p+2r-1}, N_p) \to K^{p+q+1}(N_{p+r-1}, N_p)\right]$$

Now since $K^{p+q}(N_{p+r-1}, N_{p-r}) \supseteq \ker\left[K^{p+q}(N_{p+r-1}, N_{p-r}) \to K^{p+q}(N_{p-1}, N_{p-r})\right]$ we can restrict the domain and furthermore compose with the quotient map to get a well-defined map

$$\begin{aligned} d_r^{p,q}:\ E_r^{p,q} &\to E_r^{p+r,q-r+1} \\ [x] &\mapsto [\beta \circ \delta^*_{triple}(x)] \end{aligned}$$

Corollary 3.2.2 (The first page)**.** We start by analysing the first page

$$\begin{aligned} E_1^{p,q}(M) := \frac{\ker\left[K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_{p-1}, N_{p-1})\right]}{\ker\left[\underbrace{K^{p+q}(N_p, N_{p-1}) \to K^{p+q}(N_p, N_{p-1})}_{\mathrm{id}}\right]} &\cong K^{p+q}(N_p, N_{p-1}) \\ &= \tilde{K}^{p+q}(N_p/N_{p-1}) \end{aligned}$$

Now since (N_p, N_{p-1}) is a regular index pair for $\mathrm{Crit}_p(f)$ we have a flow induced homotopiy equivalentce:

$$N_p/N_{p-1} \simeq \Big(\bigcup_{w\in Crit_p(f)} D^u_\varepsilon(w)\Big)/\Big(\bigcup_{w\in Crit_p(f)} \partial D^u_\varepsilon(w)\Big) \cong \bigvee_{w\in \mathrm{Crit}_p(f)} S^p$$

Here, we use the definitions from above

$$D^u_\varepsilon(w) \ := \ \varphi_w^{-1}\big(\{v \in T^u_w M \mid \|v\| \le \varepsilon\}\big),$$

and the last isomorphism is induced by orientations in the unstable manifolds. We can continue our inspection:

$$\begin{aligned} E_1^{p,q} &\cong \tilde{K}^{p+q}(N_p/N_{p-1}) \\ &\cong \bigoplus_{w\in\mathrm{Crit}_p(f)} \tilde{K}^q(pt) = C^k(M, f, g, \mathfrak{o}; \tilde{K}^q(pt)) \end{aligned}$$

In sum we have the chain of horizontal isomorphisms:

$$\begin{tikzcd}
E_1^{p,q} \arrow[r, "\alpha"] \arrow[d, "d_1^{p,q}"] & \tilde{K}^{p+q}(N_p/N_{p-1}) \arrow[r, "\beta"] \arrow[d] & \bigoplus_{w\in\mathrm{Crit}_p(f)} \tilde{K}^q(pt) \arrow[d] & C^p(M,\mathfrak{A},\tilde{K}^q(pt)) \arrow[l] \arrow[d] \\
E_1^{p+1,q} \arrow[r, "\alpha'"] & \tilde{K}^{p+1+q}(N_{p+1}/N_p) \arrow[r, "\beta'"] & \bigoplus_{w\in\mathrm{Crit}_{p+1}(f)} \tilde{K}^q(pt) & C^{p+1}(M,\mathfrak{A},\tilde{K}^q(pt)) \arrow[l]
\end{tikzcd}$$

By naturality of the triple boundary operator, that the second vertical map is the boundary operator. Now on the next rug we can induce two maps. From the left to make the diagram commute and from the right. Do they agree? Or asked differently, is the map induced from the d_1 map the boundary operator of morse homology?

The Main statement now is that the connecting homomorphism of K-Theory and of the long exact sequence given by the special Filtration in Morse Theory are comparable:

Theorem 3.2.3. *Given the filtration $N_{-1} \subseteq N_0 \subseteq \cdots \subseteq N_{m-1} \subseteq N_m = M$ from (8). The following diagram commutes:*

$$\begin{tikzcd}
C^k(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[r, "\partial^p"] \arrow[d] & C^{k+1}(M,\mathfrak{A},\tilde{K}^l(pt)) \arrow[d] \\
\bigoplus\limits_{p\in\mathrm{Crit}_k(f)} \tilde{K}^{k+l}(N_p,L_p) \arrow[r, "\Delta_{k+l}"] \arrow[d] & \bigoplus\limits_{q\in\mathrm{Crit}_{k+1}(f)} \tilde{K}^{k+l+1}(N_q,L_q) \arrow[d] \\
K^{k+l}(N_k,N_{k-1}) \arrow[r, "\delta_{triple}"] & K^{p+l+1}(N_{k+1},N_k)
\end{tikzcd}$$

Proof Strategy:
We will first build up a diagramm that encapusates the connecting homomorphism from K-theory, which was build as follows: Assume that (X,Y,Z) is a topological pair. Then the first map is induced from the kollaps: $Y \to Y\big/Z$. Afterwards we have the connecting homomorphism of the pair:

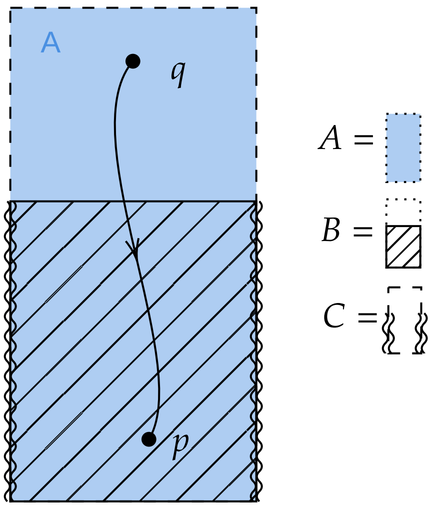
Then the K-groups of the pointed spaces $X\big/X$ and $X \cup CY$, where the latter denotes a cone above Y agree. From the latter we can quotien out X, and $X \cup CY\big/X \cong SY$. Hence we have the triple connecting homomophism as the composition:

$$K^{-1}(Y,Z) \to K^{-1}(Y) = K(SY) \cong K(X \cup CY\big/X) \to K(X \cup CY) \cong K(X\big/Y)$$

The triple connecting homomorphism for lower K-Groups is defined exactly the same via the identification $K^{-n}(X,Y) = K(S^n \vee (X,Y))$. For higher K-Groups we apply the natural Bott homomorphism. We first proof it here for the lower groups and assume k,l to be negative. Furthermore we assume for now that $l = 0$. If l is even, we can apply the Bott isomorphism and else the groups are 0. Now we choose a triple (A, B, C) in M such that:

(a) (A, B) is a regular index pair for a critical point $q \in \mathrm{Crit}_{k+1}$.

(b) (B, C) is a regular index pair for a critical point $p \in \mathrm{Crit}_k$.



From the perspective of K-theory a regular index pair for a critical point is the same as a sphere with the dimension beeing the index of said critical point. To see this just use a small index pair in a Morse chart (they are all homotopic anyway), collaps the stable part and quotiend by the exit set. By doing this we get the communative diagramm:

$$\begin{tikzcd}
K^{k-1}(B,C) \arrow[r, "\delta_{triple}"] \arrow[d] & K^k(A,B) \arrow[d] \\
K^{k-1}(S^{k-1}) \arrow[r] \arrow[d] & K^k(S^{k-1}) \\
K^k(S^k) \arrow[ur] &
\end{tikzcd}$$

And as a map between K Groups of spheres, we know that this map corresponds to the question of orientation. Now we have the maps

(a) $S^k \to S \vee (B/C)$ induced from an orientation in the unstable manifold arround p together with $-\mathrm{grad}(f)(x_i)$ where $x_i \in W(q \to p)$.

(b) $S^k \to (A/B)$ induced from an orientation in the unstable manifold arround q.

And hence the diagram

$$\begin{tikzcd} K^k(S \vee (B,C)) \arrow[rr, "\delta_{triple}"] & & K^k(A,B) \\ & K^k(S^k) \arrow[ul] \arrow[ur] & \end{tikzcd}$$

The map now factors over The K-Group of a bouquette of k-spheres generated over the connecting flow lines. In each such sphere the comparison of orientations happens.

Bevore we start proving this we need the lemma:

Lemma 3.2.4. *We work in a compact, riemanian, orientable, smooth Manifold (M,g) of dimension m together with a Morse-Smale funktion*

$$f : M \to \mathbb{R} \tag{9}$$

We assume that q is a critical point of index $k+1$ and p is a critical point of index k. we define $f(q) = b$ and $f(p) = a$ and assume that there is no critical point in $f^{-1}(a,b) \subseteq M$. We define the sub and superlevelsets

$$M^t := \{x \in M | f(x) \le t\} \quad \text{and} \quad M_t := \{x \in M | f(x) \ge t\}\,, \tag{10}$$

and the constants

$$c \in (a,b) \quad , \quad \varepsilon > 0 \text{ small} \quad , \quad T > 0 \text{ big} . \tag{11}$$

With this we define the sets:

$$N_q := \{x \in M_c | f(\varphi_{-T}(x)) \le b + \varepsilon\}\,, \tag{12}$$
$$L_q := \{x \in N_q | f(x) = c\}\,, \tag{13}$$
$$N_p := \{x \in M^c | f(\varphi_T(x)) \ge a - \varepsilon\}\,, \tag{14}$$
$$L_p := \{x \in N_p | f(\varphi_T(x)) = a - \varepsilon\}\,, \tag{15}$$

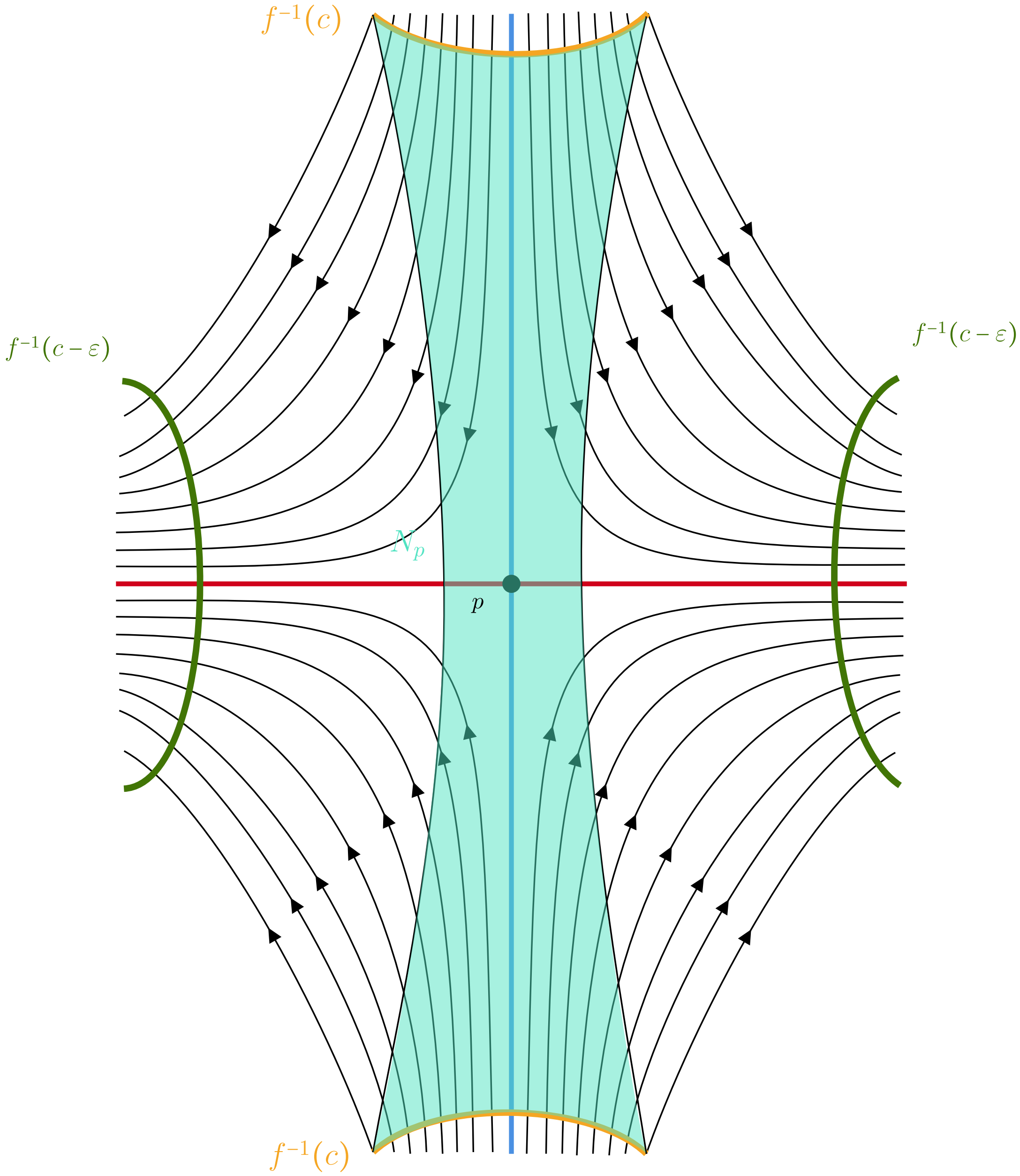
and finally:

$$C := N_p \cup N_q\,, \tag{16}$$
$$B := N_p \cup L_q\,, \tag{17}$$
$$A := L_p \cup (L_q \overset{\circ}{-} N_p)\,. \tag{18}$$

We claim that

1. (N_q, L_q) *is a regular index pair for* q .
2. (C, B) *is an index pair for* q .
3. (N_p, L_p) *is a regular index pair for* p .
4. (B, A) *is an index pair for* p .
5. N_p *is a tubular neighbourhood of* $W(\to p) \cap M^c$.

The statements 1-4 are easy to proof by the definitions. The regularity can be derived from a general fact for pairs (X, Y) in metric spaces: If Y is closed in X and there is a neighbourhood of Y that is open in X such that Y is a strong deformation retract of U. So the only thing left to show is, that N_p is a tubular neighbourhood. in [**MorseTheorySalmbon**] in the proof of lemma 3.2 Salamon claims this to be true without a proof (page 119 in the attached source). Similar, in [**banyaga2004lectures**] Banyaga claims this (also without any argument). I find this difficult to proof, since we cannot use the flow for the map from the normal bundle, as points leave N_p along the flow: The Set N_p is sketched in the figure below.



So to proof this, we first formulate two lemmas. First we restate/ refine the λ-lemma:

parameter dependence einfügen

Lemma 3.2.5 (The Technical λ–Lemma)**.** *Let M be a smooth manifold and let p be a hyperbolic fixpoint of a diffeomorphism φ on M with $\dim W(p \to) = r$ for $0 < r < m = \dim M$. Let N be an injective immersed submanifold of M that is invariant under φ and has a point of transverse intersection q with $W(\to p)$. Then for any $\varepsilon > 0$, any $r-$cell neighbourhood D of q in N that also intersects*

$W(\to p)$ transversal and any r − cell neighbourhood B of p in $W(p\to)$) there is a smaller cell $\tilde{D} \subset D$, an $n \in \mathbb{N}$ such that the following is true:

We can assume that B and $\varphi^n(\tilde{D})$ lives in a chart centered at p with a product structure $U \cong W(\to p)\cap B_\mu(0)\times W(p\to)\cap B_\mu(0)$. Here,the invers of the projection into $B \subseteq W(\to p)$:

$$h : B \to \varphi^n(\tilde{D})$$

satisfies:

1. $\sup_{x\in B} \|x - h(x)\| < \varepsilon$ *and*
2. *For the inclusions $i_1 : B \to M$ and $i_2 : \varphi^n(\tilde{D}) \to M$ we have that*

$$\sup_{x\in B} \|\mathrm{d}(i_1)_x - \mathrm{d}(i_2 \circ h)_x\| < \varepsilon\,.$$

Proof. The proov is essentially the same as given in David Hurtubise - Lectures on Morse Homology. I omitted it here. □

Next we proove that fiberwise the flow is monotonly decreasing in any normal-direction with respect to $W(\to p)$:

Lemma 3.2.6. *Let U be a tubular neighbourhood of $W(\to p)\cap M^c$, with $J : NW(\to p)\cap M^c \to M$ beeing the corresponding embedding. For $x \in W(\to p)\cap M^c$ we denote U_x the embedded fiber over x. Then there exists a $T > 0$ big enough and a global trivialization $\Phi : NW(\to p)\cap M^c \to (W(\to p)\cap M^c)\times \mathbb{R}^{\lambda_p}$ such that the the composition $f\circ\varphi_T\circ J\circ\Phi^{-1}$ is fibrewise monotonically decreasing meaning that for any $0 \neq v \in U_x$ and $\nu > 0$ small the function*

$$\begin{aligned} l_v : (1-\nu, 1+\nu) &\to \mathbb{R} \\ \lambda &\mapsto f\circ\varphi_T\circ J\circ\Phi^{-1}(x, \lambda\cdot v) \end{aligned}$$

has negative derivative.

Proof. We will apply the technichal λ-lemma. For this we reuse the notation as follows. The lemma tells us, that for any $\mu_T > 0$ we can find a T big enough, such that $\varphi_T(J(U_x))$ is close in the following sense: There is a neighbourhood L of p and a Morse chart

$$\Psi : L \to \mathbb{R}^{\lambda_p}\times\mathbb{R}^{m-\lambda_p}$$

such that $\Psi|_{\mathbb{R}^{\lambda_p}} \supseteq W(p\to)$. In this chart the set $\varphi_T(J(U_x))$ is the graph of a funktion $h^s_x : \mathbb{R}^{\lambda_p} \to \mathbb{R}^{m-\lambda_p}$ that satisfies:

$$\begin{aligned} \|\mathrm{d}h^s_v\| &< \mu_T \\ \|h^s_v\| &< \mu_T \end{aligned}$$

We define the map $h_x = h_x^s \oplus \mathrm{id}_{\lambda_p} : \mathbb{R}^{\lambda_p} \to \mathbb{R}^m$.

Now $\varphi_T(J(U_x)) = \Gamma(h_x^s) = \mathrm{im}(h_x)$. Using this we define our global trivialisation:

$$\begin{aligned}\Phi : NW(\to p) \cap M^c &\to (W(\to p) \cap M^c) \times \mathbb{R}^{\lambda_p}\\ \xi &\mapsto \left(\pi(\xi), h_{\pi(\xi)}^{-1} \circ \Psi \circ \varphi_T \circ J(\xi)\right)\end{aligned}$$

In this trivialization the funktion l_v looks like:

$$\begin{aligned}l_v(\lambda) &= f \circ \varphi_T \circ J \circ \Phi^{-1}(x, \lambda v)\\ &= f \circ \varphi_T \circ J \circ J^{-1} \circ \varphi_{-T} \circ \Psi^{-1} \circ h_x\\ &= f \circ \Psi^{-1} \circ h_x\,.\end{aligned}$$

In coordinates this reads:

Now we can calculate its derivative to conclude the statement:

$$\begin{aligned}\mathrm{d}(l_v)(\lambda) &= \mathrm{d}(f \circ \psi^{-1})\big|_{h_x(\lambda v)} \circ \mathrm{d}h_x|_{\lambda v}(v)\\ &= \mathrm{d}(f \circ \psi^{-1})\big|_{h_x(\lambda v)}(v, \mathrm{d}h_x^s(v))\\ &= -2\lambda \sum_{i_1}^{\lambda_p} (v^i)^2 + (h_x^s(\lambda v))^T \cdot (\mathrm{d}h_x^s)^i(\lambda v)\\ &\le -2\lambda \|v\|^2 + \lambda^2 (\mu^T)^2 \|v\|^2\\ &= \left(-2 + (\mu^T)^2 \lambda\right) \lambda \|v\|^2 < 0 \text{ for all } v \neq 0,\ \lambda \in I \text{ and for all } T \text{ big enough.}\end{aligned}$$

□

Hence, the sets $N_p \cap U_x$ are starshaped with respect to the embedding of the normal space.

Now we want to **smoothly** rescale to the starshaped neighbourhood. This is in general not possible and homeomorphisms from starshaped neighbourhoods to R^n are hard to describe, even if they are not required to be a resacling. The main problem in the general case is, that the following: for each $v \in \varphi_T(J(U_x))$ by the star shaped property there is a unique λ_v , such that $f \circ (\lambda_v v) = a - \varepsilon$. But the map $v \mapsto \lambda_v$ needs not to be continouos in the general starshaped setting however here it is and it even continouosly depends on x:

Definition 3.2.7 (Hyperspherical coordinates)**.** For our next proof we need to introduces spherical coordinates on R^{λ_p}. Define $s_k := \sin(\theta_k)$ and $c_k := \cos(\theta_k)$:

$$\Lambda : \mathbb{R}_{\geq 0} \times [0,\pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}^{\lambda_p}$$
$$(r, \theta_1, \ldots, \theta_{\lambda_p - 1}) \mapsto \begin{pmatrix} rc_1 \\ rs_1c_2 \\ \vdots \\ rs_1 \cdots s_{\lambda_p-2}c_{\lambda_p-1} \\ rs_1 \cdots s_{\lambda_p-2}s_{\lambda_p-1} \end{pmatrix}$$

With differential

$$\begin{pmatrix} c_1 & -rs_1 & 0 & 0 & \cdots & 0 \\ s_1c_2 & rc_1c_2 & -rs_1s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \ddots & \vdots \\ & & & & & \\ s_1\cdots s_{n-2}c_{n-1} & \cdots & \cdots & & & -rs_1\cdots s_{n-2}s_{n-1} \\ s_1\cdots s_{n-2}s_{n-1} & rc_1\cdots s_{n-1} & \cdots & & & rs_1\cdots s_{n-2}c_{n-1} \end{pmatrix}$$

We have shown in lemma 3.2.6 that the maps l_v are monotonically decreasing. Hence the condition $f(x) \geq a - \varepsilon$ gives rise to a starshaped fiber. But we want the fibers to be homeomorphic to R^{λ_p} and the dream would be to just rescale it as follows: Assume that $\rho > 0$ is small enough, such that for all $x \in (W(\to p) \cap M^c)$ we have that $\sup_{x \in h_x(B_\rho(0))} (f(x)) \geq a - \varepsilon$. Now we have the maps:

f mit $f \circ \Psi^{-1}$ ersetzen, da hier eigentlich die Morse karte genutzt wird!!

$$r_x : h_x(\partial(B_\rho(0))) \to \mathbb{R}_{\geq 1}$$
$$v \mapsto \lambda_v \text{ such that } f(\lambda_v v) = a - \varepsilon$$

Notice that for small enough ε such a λ_v exists and the lemma 3.2.6 gives us uniqueness. Now we want to have continouoity and furthermore continouos dependence on x for which we will use the parameter-dependent implicit funktion theorem:

Lemma 3.2.8. *The parameterdependent funktions r_x are continouos and depend continouos on x*

Proof. We redefine the funktion r_x by using spherical coordinates as follows: Implizitly for each x we have the Mapping:

$$F_x : \mathbb{R}_{\geq 0} \times [0,\pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}$$
$$(r, \theta_1, \ldots, \theta_{\lambda_p - 1}) \mapsto f \circ h_x \circ \Lambda(r, \theta_1, \ldots, \theta_{\lambda_p - 1})$$

Here the funktion reads:

$$f \circ h_x \circ \Lambda(\underbrace{r, \theta_1, \dots, \theta_{\lambda_p - 1}}_{:=v}) = f(\begin{pmatrix} h_x^s(\Lambda(v)) \\ \Lambda(v) \end{pmatrix}) = -2r^2 + 2 \sum_{i=\lambda_p+1}^{m} (h_x(\Lambda(v)))_i^2$$

To make use of the implizit funktion theorem, we inspect the differential in r-direktion:

$$\frac{\partial}{\partial r}(-2r^2 + 2 \sum_{i=\lambda_p+1}^{m} (h_x(\Lambda(v)))_i^2) = -4r + 2 \sum_{i=\lambda_p+1}^{m} \left(\sum_{j=1}^{\lambda_p} \left(\frac{\partial (h_x)_i}{\partial x_j} \frac{\partial \Lambda_j}{\partial r} \right) \right)$$

This does not vanish for big enough r, since the partial differentials of λ are of norm one, and those of h_x are small. Hence for big enough r , lets say $r \geq \mu > 0$ this is negative and thereby we get a funktion

$$\tilde{r_x} : [0, \pi]^{\lambda_p - 2} \times [0, 2\pi) \to \mathbb{R}_{\geq \mu}$$

such that $(f \circ h_x \circ \Lambda)^{-1}(a - \varepsilon)$ is the graph of $\tilde{r_x}$. Hence, if we define $\pi_r : \mathbb{R}_{\geq 0} \times [0, \pi]^{\lambda_p - 2} \times [0, 2\pi) \to [0, \pi]^{\lambda_p - 2} \times [0, 2\pi)$ to be the projection we have that

$$\begin{aligned} r_x : h_x\left(\partial B_\rho(0)\right) &\to \mathbb{R} \\ v &\mapsto \tilde{r}_x \circ \pi_r(\Lambda^{-1}(h_x^{-1} v)) \end{aligned}$$

is continouos and depends continouosly on x. (In reality we still need to check for smoothness in $\Lambda(\{\theta_{\lambda_p - 1} = 0\}$. This can be done by a different parametrisation of the sphere meaning by different spherical coordinates.)The resulting funktion agrees with the earlyer definition of r_x concluding the proof. □

Proof That N_p is Tubular. To finish our proof we rescale the Fibers smoothly. For this we first restrict the domain of J: Let $\nu > 0$ be small enough, such that for

$$N_\nu W(\to p) \cap M^c := \{\xi | \text{ if } \Phi(\xi) = (x, v), \|v\| < \nu\}$$

the supremum

$$\sup_{x \in \varphi_T \circ J(N_\nu W(\to p) \cap M^c)} \geq a - \varepsilon$$

Hence, $J(N_\nu W(\to p) \cap M^c)$ is included in N_p and now we resscale the fibres $U_x^\nu := U_x \cap \mathrm{im} J(N_\nu W(\to p) \cap M^c) \supseteq U_x$ to get a surjective inclusion into N_p. For all $x \in W(\to p) \cap M^c$ we define the smooth rescaling as follows:

$$\begin{aligned} RS : U_x^\nu &\to U_x \\ v &\mapsto \varphi_{-T} \circ \Psi^{-1} \circ r_x \left(\rho \cdot \frac{h_x^{-1} \circ \Psi \circ \varphi_T(v)}{\|h_x^{-1} \circ \Psi \circ \varphi_T(v)\|} \right) \end{aligned}$$

This rescaling is smooth, smoothly depends on x, and it maps surjectifly onto $N_p \cap U_x$. □

The same arguments work to show that N_q is a tubular neighbourhood of $W(q \to) \cap M_c$, by replacing f with $-f$. Furthermore, we can assume, that the V_j from below are fibers in the normal neighbourhood. This can be realized, since V_j intersect transversal and by trivialization we have a global fram of the bundel and hence can smoothly rescale the embedding along the frame to get that V_j is the image of the fibre above x_j.

Proof of Theorem 3.2.3. Lets say that $p \in -\mathbb{N}$ is negative for the moment. By definition we get the triple connecting morphism as the composition of the inclusion with the connecting homomorphism of the pair:

$$\begin{tikzcd}
& K^p[N_q \cup N_p, L_q \cup N_p] \arrow[r, Rightarrow] & K^p[N_q \cup N_p \cup /(L_q \cup N_p)] \\
& & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p)] \arrow[u, "(m^*)^{-1}"] \\
K^{p-1}[L_q \cup N_p] \arrow[uur, "\delta_{pair}"] \arrow[r, Rightarrow] & K^p[S_1 \wedge (L_q \cup N_p)] \arrow[r, "\alpha^*"] & K^p[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \arrow[u, "i^*_\alpha"] \\
K^{p-1}[L_q \cup N_p, L_p \cup (L_q \setminus \mathring{N}_p)] \arrow[u, "i^*"] \arrow[r, Rightarrow] & K^p[S_1 \wedge L_q \cup N_p, S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p)] \arrow[u, "i^*_\beta"] &
\end{tikzcd}$$

The bottom left diagramm commutes by definition. All maps with names $i^*_{something}$ are induced from obvious inclusions. The other maps are defined as follows, where p denotes the choosen point of our pointed spaces:

$$\alpha : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) \to S_1 \wedge (L_q \cup N_p)$$
$$x \mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \\ x & \text{else} \end{cases}$$

This map is well defined and continouos, as every point in $S_1 \wedge (L_q \cup N_p)$ is of the form $[(t_1, x)]$ where $x \in L_q \cup N_p$. and those are the points in the domain of α that not get collapsed to p We have the maps:

$$c_\gamma : [N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)] \to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right]$$
$$x \mapsto \begin{cases} p & \text{if } x \in C_2(N_q \cup N_p) \quad \cup \quad C_1(L_p \cup \overline{(L_q \setminus N_p)}) \\ x & \text{else} \end{cases}$$

and

$$c_\beta : [S_1 \wedge (L_q \cup N_p)] \to \left[S_1 \wedge L_q \cup N_p \Big/ S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \right]$$
$$x \mapsto \begin{cases} p & \text{if } x \in S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p) \\ x & \text{else} \end{cases}$$

This map is again well defined and continouos and is also a map of pairs. In fact, both maps α and β can be written as $x \mapsto \overline{x}$ and since the collapsed space is contractible, they induce isomorphism between the K-groups. The map γ is just the identity.

Furthermore, we have the commutative diagramm:

$$\begin{tikzcd}
K^p\big[S_1 \wedge (L_q \cup N_p)\big] \arrow[r, "\alpha^*"] & K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)\right] \\
K^p\Big[{}^{S_1 \wedge L_q \cup N_p}\Big/_{S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p)} \Big] \arrow[u, "c_\beta^*"] \arrow[r, "\gamma^*"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right] \arrow[u, "c_\gamma^*"']
\end{tikzcd}$$

To see the commutativitie we have to show that

$$c_\beta \circ \alpha = \gamma \circ c_\gamma : N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p) \to \left[{}^{S_1 \wedge L_q \cup N_p}\Big/_{S_1 \wedge L_p \cup (L_q \setminus \mathring{N}_p)} \right]$$

since α and β are both of the form $x \mapsto \overline{x} = x$ or p we compare the case distinctions:

- $c_\beta \circ \alpha(x) = p \Leftrightarrow x \in C_1\left(L_p \cup (L_q \setminus \mathring{N}_p)\right) \cup C_2(N_q \cup N_p)$
- $\gamma \circ c_\gamma(x) = p \Leftrightarrow x \in C_2(N_q \cup N_p) \cup C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)$

Which are the same, since $L_q \setminus \mathring{N}_p = \overline{L_q \setminus N_p}$ due to L_q being closed. Now we consider the maps:

$$c : [N_q \cup N_p \cup C_1(L_q \cup N_p)] \to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\mathrm{cl}\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)} \right]$$

$$x \mapsto \begin{cases} p & \text{if } x \in \mathrm{cl}(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j) \\ x & \text{else} \end{cases}$$

and

$$c_\iota : [N_q \cup N_p \cup C_1(L_q \cup N_p)] \to \left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{N_q \cup N_p \cup C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \right]$$

$$x \mapsto \begin{cases} p & \text{if } x \in N_q \cup N_p \cup C_1(L_p \cup \mathrm{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p) \\ x & \text{else} \end{cases} .$$

We define V_j to be the the finitly many components $N_p \cap S(q \to)$ and hence we can identify the sets $\overline{L_q \setminus N_p} = \mathrm{cl}(L_q \setminus (L_q \cap N_q)) = \mathrm{cl}(L_q \setminus (\bigcup_{j=1}^{l}))$ and with this the map η in the diagramm below is just the identity:

$$\begin{tikzcd}
K^p\left[N_q \cup N_p \cup \big/ (L_q \cup N_p)\right] & \\
K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] \arrow[u, "(m^*)^{-1}"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p)}{\operatorname{cl}\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[l, "c^*"'] \\
K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)\right] \arrow[u, "i_\alpha^*"] & K^p\left[\frac{C_1(L_q \cup N_p)}{\operatorname{cl}\left(C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[u, "i_\delta^*"] \\
 & K^p\left[\frac{C_1(L_q)}{\operatorname{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \arrow[u, "i_\eta^*"] \\
K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)}\right] \arrow[uu, "c_\gamma^*"] & K^p\left[\frac{N_q \cup N_p \cup C_1(L_q \cup N_p) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)} \Big/ \frac{N_q \cup N_p \cup C_1(L_p \cup \operatorname{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)}{C_2(N_q \cup N_p)}\right] \arrow[u, "c_\iota^*"] \arrow[l, "\eta^*"]
\end{tikzcd}$$

Now the mamoth task is to show that this beast commutes: For this we preoceed with the same strategy since again all maps are inclusions and collases and hence the maps look like $x \mapsto \overline{x} = x$ or p. For the consideration assume that $x \neq p$ in the domain. Then:

$$\eta \circ c_\gamma \circ i_\alpha(x) = p \Leftrightarrow x \in C_1(L_p \cup \overline{(L_q \setminus N_p)}) \cup C_2(N_q \cup N_p)$$

On the other hand:

$$\begin{aligned}
&c_\iota \circ i_\eta \circ i_\delta \circ c(x) = p \Leftrightarrow c(x) = p \text{ or } c_\iota(x) = p \\
x \in \operatorname{cl}&\left(N_q \cup N_p \cup C_1(L_q \cup N_p) \setminus \bigcup_{j=1}^{l} C_1 V_j\right) \cup N_q \cup N_p \cup C_1(L_p \cup \operatorname{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p) \\
&= C_1 \operatorname{cl}\left((L_q \cup N_p) \setminus \bigcup_{j=1}^{l} V_j\right) \cup C_1(L_p \cup \operatorname{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p) \\
&= C_1 \operatorname{cl}\left((L_q) \setminus \bigcup_{j=1}^{l} V_j\right) \cup C_1(L_p \cup \operatorname{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p) \\
&= C_1(L_p \cup \operatorname{cl}\left(L_q \setminus \bigcup_{j=1}^{l} V_j\right)) \cup C_2(N_q \cup N_p)
\end{aligned}$$

And now the conditions agree and hence the big diagramm commutes. Now notice how we have the homeomorphism

$$\Lambda : \left[\frac{C_1(L_q)}{\operatorname{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1 V_j\right)}\right] \to \bigvee_{j=1}^{l} C_1 V_j \big/ \partial C_1 V_j$$

that is the identity on all $C_1\mathring{V}_j$. Furthermore we have the compression maps

$$c_j : \bigvee_{j=1}^{l} C_1V_j/\partial C_1V_j \to C_1V_j/\partial C_1V_j$$

Hence by the additivity of K–Theorie, the pair

$$\left(K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q) \setminus \bigcup_{j=1}^{l} C_1V_j\right)}\right], \left((c_j \circ \Lambda)^*\right)_j \right)$$

satisfies the coproduct universal property.

So consider the map

$$\kappa : c^* \circ i_\delta^* \circ i_\eta^* \circ (c_j \circ \Lambda)^* : K^p\left[C_1V_j/\partial C_1V_j\right] \to K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] \tag{19}$$

This map is induced from

$$\begin{aligned} N_q \cup N_p \cup C_1(L_q \cup N_p) &\to C_1V_j/\partial C_1V_j \\ x &\mapsto \begin{cases} x & \text{if } x \in C_1(\mathring{V}_j)\,, \\ p & \text{else .} \end{cases} \end{aligned}$$

For all C_1V_j we need an homeomorphism to S^{k+1}:

$$\begin{aligned} r : C_1V_j/\partial C_1V_j &\to I \times D^k/\partial(I \times D^k) \\ (t, x) &\mapsto (t, \psi(x)) \end{aligned}$$

For this we use that N_p is a tubular neighbourhood of $W(\to p) \cap M^c$ where that latter is contractable. Hence, we have a homotopy equivalence by contracting the base space and the fibers simultaniouosly via a frame. and the transversality ensures, that this gives a homeomorphism from V_j to the fibre over p, which is a ball $B \subseteq W(p \to)$. and composed with the refined Morse chart this is a Ball B^{λ_p} in $\mathbb{R}^{\lambda_p}$. This gives us a well defined homeomorphism:

$$r : C_1V_j/\partial(C_1V_j) \to C_1B^{\lambda_p}/\partial(C_1B^{\lambda_p})$$

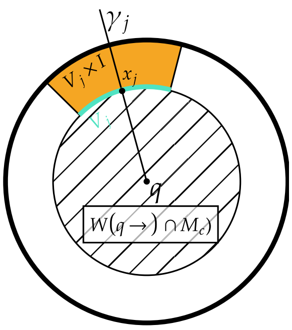
Hence, an orientation on $TW(p \to)$, induces a generator α of $K^p\left[C_1V_j/\partial(C_1V_j)\right]$. Here the orientation is introduced by choosing the refined Morse chart.

On the other hand, we have a generator on $N_p \cup N_p \cup C_1(L_p \cup N_p)$ induced from an orientation on $TW(q \to)$ as follows: First we get a generator of the K group of the Regular index pair $(N_q/L_q) \simeq (N_q \cup N_p/L_q \cup N_p)$ by choosing a small index pair and the homotopy. This generator induces a generator β of $K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right]$ by pulling it back vie the contratcion of the cone C_1. Now we need to study, how the image of α under the map (19) relates to β as seen in the diagram.

$$\begin{tikzcd}
 & K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] & \\
 & K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q)\setminus\bigcup_{j=1}^{l} C_1V_j\right)}\right] \arrow[u] & K^p\left[C_1V_j/\partial C_1V_j\right] \arrow[l] \arrow[ul, "\kappa"'] \\
K^p\left[N_q/(L_q)\right] \arrow[r] & K^p[D^u_\varepsilon/\partial D^u_\varepsilon] \arrow[r] & K^p[S^{\lambda_p+1}] \arrow[d] \\
K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] \arrow[u] \arrow[ur] & & K^p\left[C_1T^u_pM/\partial C_1T^u_pM\right] \arrow[dl] \\
K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q)\setminus\bigcup_{j=1}^{l} C_1V_j\right)}\right] \arrow[u] & K^p\left[C_1V_j/\partial C_1V_j\right] \arrow[l] \arrow[ul, "\kappa"'] &
\end{tikzcd}$$

Hier ist noch einiges zu tun!

Proof Strategy:
The idea of the next proof is the following: an orientation in $N_{x_j}\gamma_j$ induces a generator of the k-th K group $C_1V_j/\partial(C_1v_j)$. This generator can be pulled back to be a generator of the k-th K group of $W(q\to)\cap M_c/\partial(W(q\to)\cap M_c)$ via the collapsing map. This pulling back coincides with the map from (19). Now assume that the basis $(b_j, -\mathrm{grad}(f))$ of $N_{x_j}\gamma_j \oplus T_{x_j}\gamma_j$ is positifly oriented with respect to the orientation of T^u_qM. And assume furthermore, that the inverse of the embedding E^u_q is an oriented chart. Then the pulled back generator agrees with the generator of $W(q\to)\cap M_c/\partial(W(q\to)\cap M_c)$ that is induced by directly using E^u_q.



For this we make use of an collar neighbourhood of $W(q\to)\cap M_c$ defined as follows: Again using the implicit funktion theorem and spherical coordinates (compare referenz) we have that the preimage of $W(q\to)\cap M_c$ under the embeddign $E^u_q : T^u_qM \to W(q\to)$ is (after rescaling the funktion) a Disk, So withour restrictions assume that $E^u_q(B_1(0)) = W(q\to)\cap M_c$. Then $\mathfrak{C} := E^u_p(B_2(0))$ is a collar neighbourhood and $V_j \times I \hookrightarrow L_i \subseteq \mathfrak{C}$ embedds in the obvious way. This is done by mapping $V_j \times I$ fisrst into $B_2(0)$ via $(v,t) \mapsto (E^u_q)^{-1}v\cdot(1+t)$ and then applying E^u_p.

This mapping E^u_q gives also rise to an oriented chart of V_j: For this we deform $B_2(0)$ to be a cylinder along γ_j by changing the base such that $(E^u_q)^{-1}x_j$ is the first base vector whilst keeping the orientation (det of base change is one). Then just strech the disc:

$$(x_1, \dots x_{\lambda_p+1}) \mapsto \left(\frac{x_1}{|x_1|}, \dots x_{\lambda_p+1}\right).$$

. By transversality of V_j and γ_j we can assume that the orthogonal projection Π_j along the first basis vector restricted to the preimage of V_j is a homeomorphism. Now we get the chart as a composition:

$$E^u_{q,j} := \Pi_j \circ (E^u_q)^{-1} : V_j \to \mathbb{R}^{\lambda_p}$$

Here is where the definition of the Morse boundary operator comes because now we have two different orientations in V_j. One induced from $T^u_p M$ (lets call the oriented basis b^u_p) and one induced from above. Hence in $Tx_j\gamma_j \oplus Tx_j V_j$ we have two different oriented basis induced from this chart and from the chart given by the tubular neighbourhood embedding of $W(\to p) \cap M^c$ restricted to the normal fiber above x_j.

$$\begin{tikzcd}
\mathbb{R}^{\lambda_p} \arrow[r, dashed, "b^u_p"] \arrow[d, "(E^u_{qj})^{-1}"'] & T^u_p M \arrow[dl, "J|_{U x_j}"] \\
V_j &
\end{tikzcd}$$

If we now swich to the Alexandrof kompakitifikations we have that this diagramm kommutes up to homotopy, if and only if the orientaions agree. . Else we have a change of sign. Hence we have the diagramm that commutes up to sign:

hier beweis nachholen. Sollte eine einfache übungsaufgabe sein mittels abbildungsgrad zwischen sphären ist homotopieinvariante

$$\begin{tikzcd}
S \vee S^{\lambda_p} \arrow[r, "\mathrm{id} \vee b^u_p"] \arrow[d, "\mathrm{id} \vee (E^u_{qj})^{-1}"'] & \frac{C_1 T^u_p M}{\partial C_1 T^u_p M} \arrow[dl, "\mathrm{id} \oplus J|_{U x_j}"] \\
\frac{C_1 V_j}{\partial C_1 V_j} &
\end{tikzcd}$$

Here the right path coincides with the path $f_2 \circ f_1$. Now we introduce a left part that will later factor over the collaps that induces κ:

$$\begin{tikzcd}
N_q C_1(L_q) \arrow[dr, bend right, "collaps"'] & S \vee S^{\lambda_p} \arrow[l, "f_5"'] \arrow[r, "\mathrm{id} \vee b^u_p"] \arrow[d, "\mathrm{id} \vee (E^u_{qj})^{-1}"'] & \frac{C_1 T^u_p M}{\partial C_1 T^u_p M} \arrow[dl, "\mathrm{id} \oplus J|_{U x_j}"] \\
& \frac{C_1 V_j}{\partial C_1 V_j} &
\end{tikzcd}$$

We define f_5 as follows:

$$f_5 : S \vee S^{\lambda_p} = \frac{B_2(0)}{\partial B_2(0)} \to N_q C_1(L_q)$$

$$x \mapsto \begin{cases} E^u_q(x) & \text{if } \|x\| \leq 1 \\ (\lambda, x_j) & \text{if } x = (\lambda + 1) \cdot x_j, \text{ where } \lambda > 0 \text{ and } x_j \in (E^u_q)^{-1}(V_j) \\ p & \text{else.} \end{cases}$$

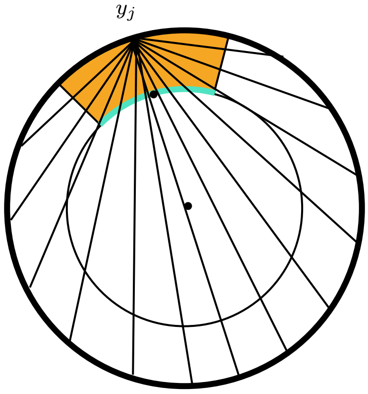
Now the left side commutes and also factos over the collaps that induces κ. hence κ is up to homotopie the plus or minus the identity which is determined by the sign of the Morse boundary operator at γ_j.

$$\begin{tikzcd}
K^p\left[N_q/(L_q)\right] \arrow[r] & K^p[D^u_\varepsilon/\partial D^u_\varepsilon] \arrow[r, "f_4"] & K^p[S^{\lambda_p+1}] \arrow[d, "f_1"] \\
K^p\left[N_q \cup N_p \cup C_1(L_q \cup N_p)\right] \arrow[u] \arrow[ur, "f_3"] & K^p[\frac{B_2(0)}{\partial B_2(0)}] \arrow[u, "b_2"] \arrow[r, "b_1"] \arrow[d, "((E^u_q)^{-1}\circ i)^*"] & K^p\left[C_1T^u_pM/\partial C_1T^u_pM\right] \arrow[dl, "f_2"] \\
K^p\left[\frac{C_1(L_q)}{\mathrm{cl}\left(C_1(L_q)\smallsetminus\bigcup_{j=1}^{l} C_1V_j\right)}\right] \arrow[u] & K^p\left[C_1V_j/\partial C_1V_j\right] \arrow[l] \arrow[ul, "\kappa"] &
\end{tikzcd}$$

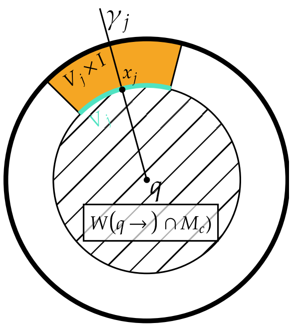
Now we start going arround that diagramm wich is commutative up to hopotopie and start at the bottom. b_1 is the same map as f_1 and hence makes use of the orientation of T^u_pM. $f_2\circ b_1$ is homotopic to the inculsion induced mapping $\left((E^u_q)^{-1}\circ i\right)^*$. This can be seen as follows: First notice how $f_2\circ b_1$ is homotopic to map induced from $(\mathrm{id}\oplus E^u_{q,j})^{-1})$

$$(x_1,\dots x_{\lambda_p+1})\mapsto(\frac{x_1}{|x_1|},\dots x_{\lambda_p+1}).$$

now the half cylinder $(x_1\geq 0)$ is again homotopic to the whole with the inclusion and collapsing being the homotopic invers maps. Those considerations also work modulo the boundary. Now $f_2\circ b_1$ is homotopic to $\left((E^u_q)^{-1}\circ i\right)^*$ if and only if the chart $(E^u_q)^{-1}$ by applying the streching map $s:\frac{B_2(0)}{\partial B_2(0)}\to\frac{B_2(0)}{\partial B_2(0)}$: (which is homotopic to the identity.)



Using this streching we have that



Furthermore, $V_j\times I$ is homeomorphic to $\mathfrak{C}$ by stre
streching can be done along rays starting in y_j =
is homotopic to the identity, if viewed as a en
in $W(q\to)\cap M_c$. We can now define the col
$W(q\to)\cap M_c$ in a continouos way by defining
identity inside of L_j and outside we map each
intersection of the ray containing said point with
of L_p. Das sollte die letzte Lücke sein!

$\square$